

Chapter 8

Background Estimation

The background prediction is obtained from the simulated samples introduced in Section 5.3, with the dominant $W + \text{jets}$ and top-quark contributions normalised to data in control regions (CRs). This chapter describes these constraints, the treatment of the small fake- τ contribution, and the validation of the prediction in regions outside the SRs. The object definitions and SR selections follow Chapters 6 and 7.

8.1 Strategy overview

The CRs provide data samples enriched in the backgrounds to be normalised, with little signal contamination. Four CRs are used: CRW-Res and CRW-NonRes constrain $W + \text{jets}$, while CRTop-Res and CRTop-NonRes constrain top-quark production. The total event count in each CR is used in a simultaneous fit. The fitted normalisation factors, λ_W^{Res} , $\lambda_W^{\text{NonRes}}$, λ_t^{Res} and $\lambda_t^{\text{NonRes}}$, multiply the corresponding simulated yields in the CRs, SRs and validation regions (VRs). The same top-quark factor is applied to $t\bar{t}$ and single-top production within each topology. Separate factors for the resonant and non-resonant categories allow the normalisation to differ between the high- and low-jet- p_T selections.

The extrapolation from a CR to an SR is supplied by simulation: the fit fixes the background rate in the CR, while the simulated ratio of SR to CR yields determines the rate transferred to the SR. The $W + \text{jets}$ extrapolation therefore depends on the description of the lepton flavour and jet flavour, and the top-quark extrapolation also depends on the acceptance of the additional lepton and the relaxed CR requirements. The relevant uncertainties are described in Chapter 9. Diboson and $Z + \text{jets}$ production, together with the other small contributions, retain their simulated nominal yields and are allowed to vary within their assigned uncertainties in the fit. The fitted uncertainty parameters are propagated to the SRs and VRs together with the

normalisation factors. The multijet suppression is already described in Section 7.1.2 and Appendix B.

The VRs test these extrapolations without contributing event counts to the background-only fit. Their object multiplicities are summarised in Table 8.1. In particular, VRW tests the change from a b -jet veto to one tagged jet, and VRTop tests the prediction for a hadronic τ with no selected light lepton. The zero- b -jet regions containing a τ are the SR0b regions defined in Section 7.3. The likelihood construction and the final fitted normalisation factors are reserved for Chapter 10.

Table 8.1 Roles and object multiplicities of the CRs and nominal VRs. Each row represents separate resonant and non-resonant regions. The supplementary $1\tau 0\ell 1b$ regions are defined in Section 8.5.4.

Region	N_τ	N_ℓ	N_b	Purpose
CRW	0	1	0	Constrain $W + \text{jets}$ normalisation
CRTop	1	1	≥ 1	Constrain top-quark normalisation
VRW	0	1	1	Test the $0b \rightarrow 1b$ extrapolation
VRTop	1	0	2	Test the top prediction without a light lepton

8.2 $W + \text{jets}$ estimation

8.2.1 Control-region definition

The CRW regions select $W \rightarrow \ell \nu$ events by replacing the τ in the corresponding SR0b selection with an electron or muon. The b -jet veto suppresses top-quark production and provides a larger $W + \text{jets}$ sample than a tagged selection. Exactly one signal electron or muon and no selected τ are required. The signal-lepton requirements are those in Chapter 6. Single-lepton triggers are used, as listed in Section 5.4. This choice avoids the different response of the calorimeter-based E_T^{miss} trigger to electrons and muons, since the trigger-level missing momentum does not include the muon momentum.

The lepton replaces the τ in the transverse mass and angular separation, giving m_T^ℓ and $\Delta\phi(\ell, E_T^{\text{miss}})$ through the definitions in Section 7.1.1. The leading jet is used for the resonant/non-resonant categorisation and the invariant mass $m_{j\ell}$. Table 8.2 gives the selections together with those of VRW. The kinematic correspondence to the SRs restricts the extrapolation to similar hard-scattering configurations. The neutrino from a hadronic τ decay changes both the visible τ momentum and $\vec{p}_T^{\text{miss}}$, so the lepton replacement does not imply identical reconstructed distributions. These differences are retained in the simulated acceptance ratio.

The extrapolation is checked in the lower-mass $W \rightarrow \tau\nu$ sample described in Section 8.5.4. Both CRs have a $W + \text{jets}$ purity of approximately 70%. Diboson production is the largest secondary contribution in CRW-Res.

Table 8.2 Selections for CRW and VRW. Both require $N_\tau = 0$, $N_\ell = 1$ and a single-lepton trigger. The common event-quality and multijet requirements are retained. The selected jet is the leading jet in CRW and the tagged jet in VRW. Its momentum is denoted by p_T^J .

Requirement	Resonant	Non-resonant
N_b	0 in CRW, 1 in VRW	
p_T^ℓ [GeV]	> 200	
p_T^J [GeV]	> 250	$50 < p_T^J < 250$
E_T^{miss} [GeV]	> 200	> 400
m_T^ℓ [GeV]	> 200	> 600
$\Delta\phi(\ell, E_T^{\text{miss}})$	No requirement	> 1.2
$m_{j\ell}$ or $m_{b\ell}$ [GeV]	> 800	No requirement
N_j	≤ 4	≤ 2

8.2.2 Pre-fit comparisons

The pre-fit comparisons in Figures 8.1 and 8.2 test the simulated distributions before the CR normalisation is fitted. The $N - 1$ convention is the one introduced in Section 7.2. The electron and muon channels are shown separately to test their consistency before combining them. The plotted bands and the uncertainties in Table 8.3 are statistical only. No clear trend in the data-to-prediction ratio is observed across the displayed kinematic distributions, although their overall normalisations differ. The raw data-to-total-MC ratios are diagnostic quantities: they are not the fitted $W + \text{jets}$ normalisation factors, since the CRs also contain other backgrounds.

Table 8.3 Observed counts and data/total-MC ratios in the CRs before fitting. The uncertainties are statistical.

Region	Electron channel		Muon channel	
	Data	Data/MC	Data	Data/MC
CRW-Res	243	1.25 ± 0.09	219	1.08 ± 0.08
CRW-NonRes	474	1.00 ± 0.09	398	0.80 ± 0.08
CRTop-Res	30	0.73 ± 0.13	33	0.86 ± 0.15
CRTop-NonRes	240	0.81 ± 0.05	228	0.83 ± 0.06

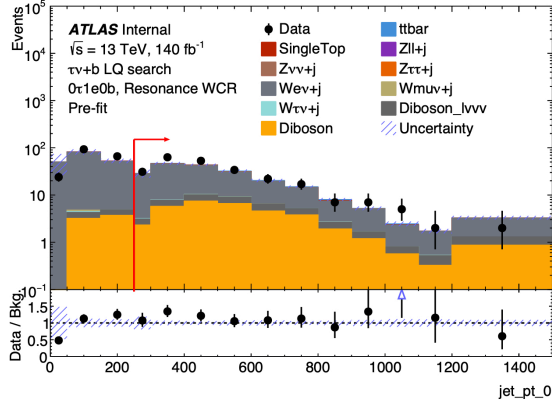
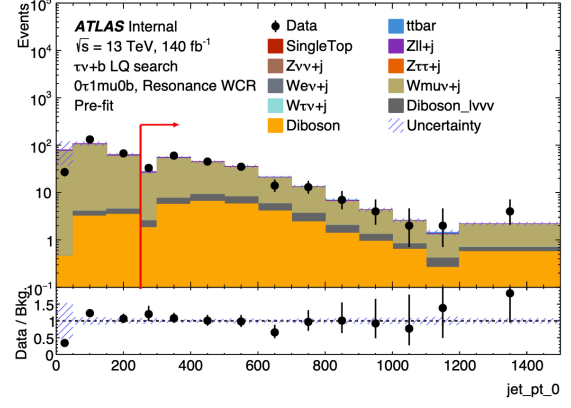
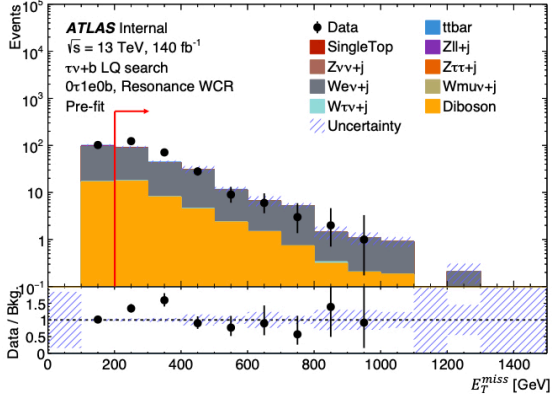
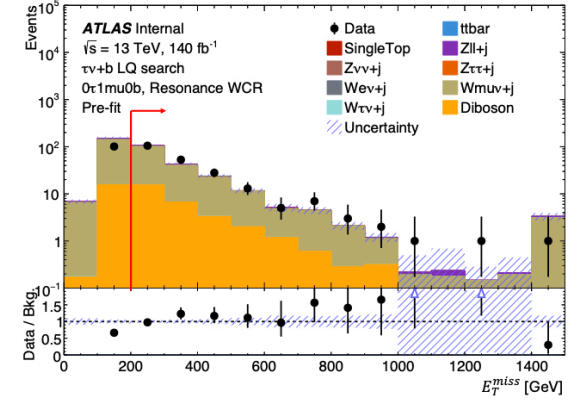
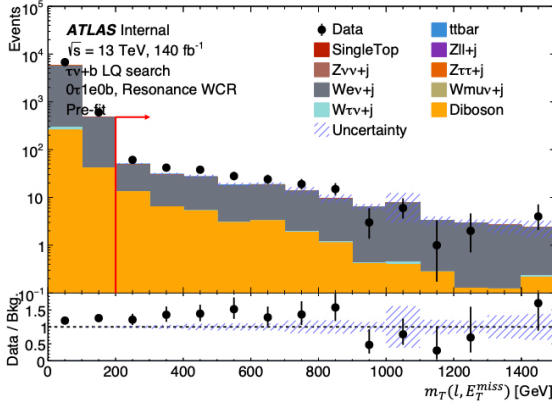
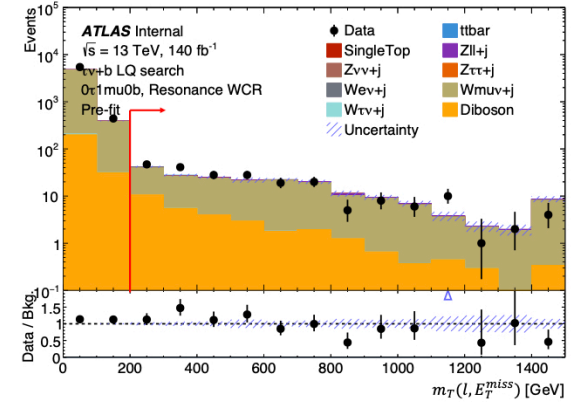
(a) p_T^{j1} , e channel(b) p_T^{j1} , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T^l , e channel(f) m_T^l , μ channel

Fig. 8.1 Pre-fit $N - 1$ distributions in CRW-Res, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

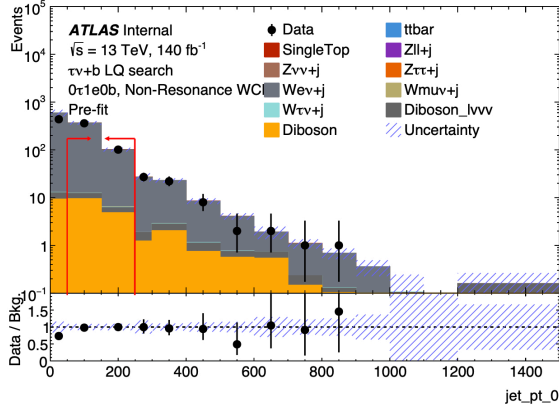
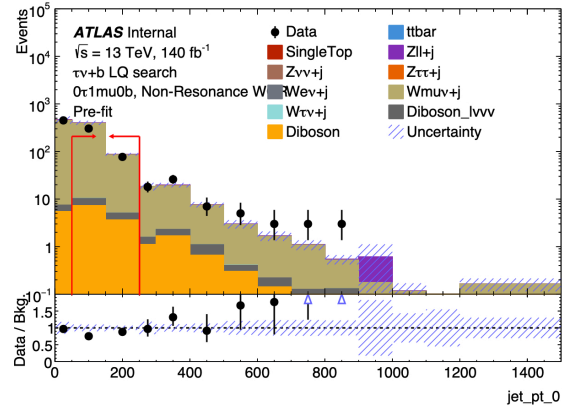
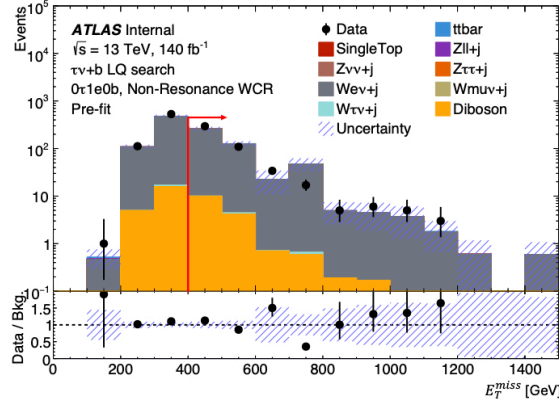
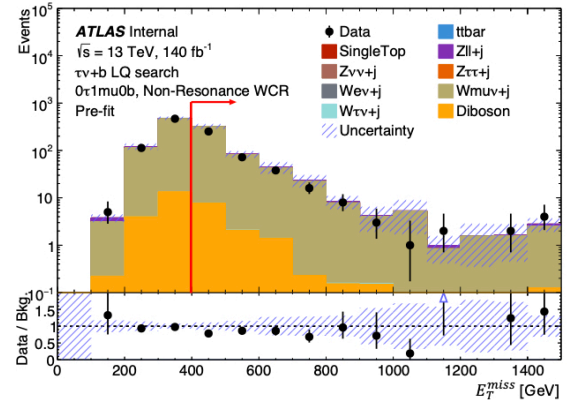
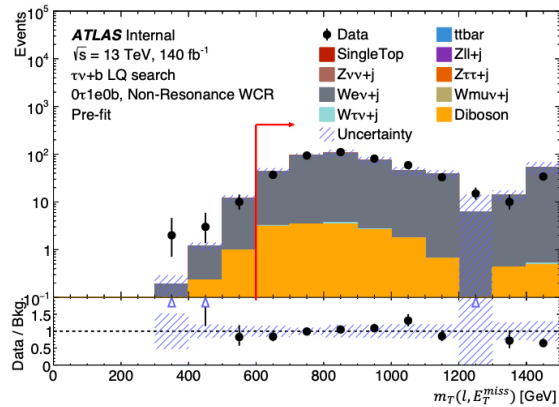
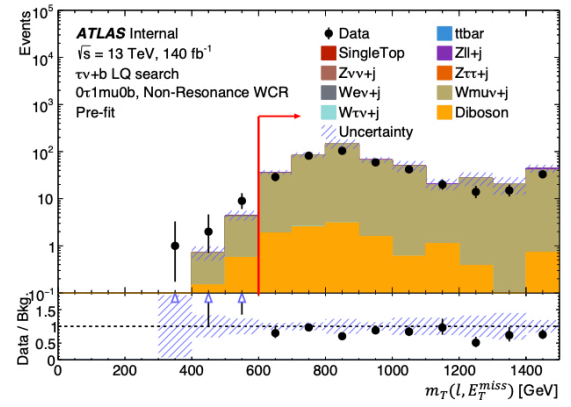
(a) p_T^{j1} , e channel(b) p_T^{j1} , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T^ℓ , e channel(f) m_T^ℓ , μ channel

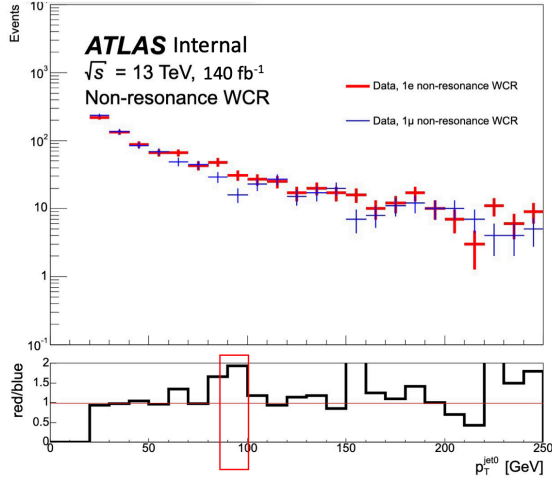
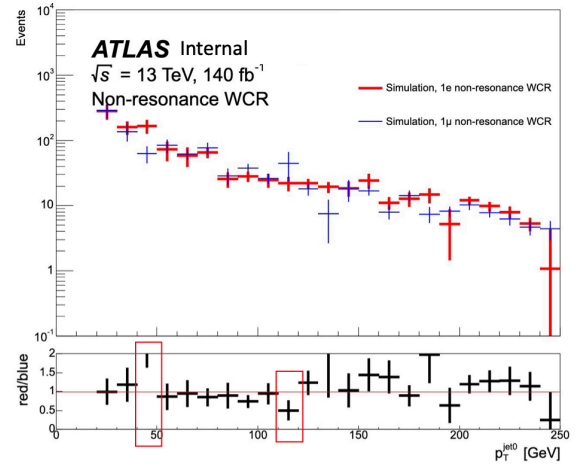
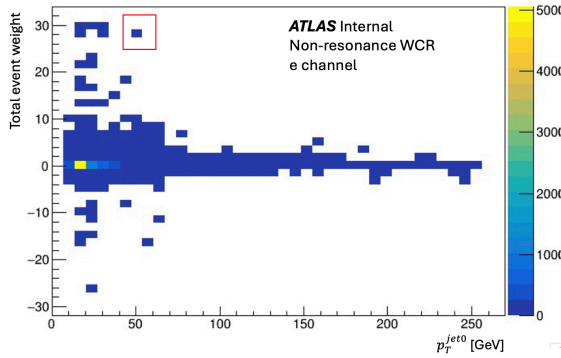
Fig. 8.2 Pre-fit $N - 1$ distributions in CRW-NonRes, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

8.2.3 Electron–muon consistency

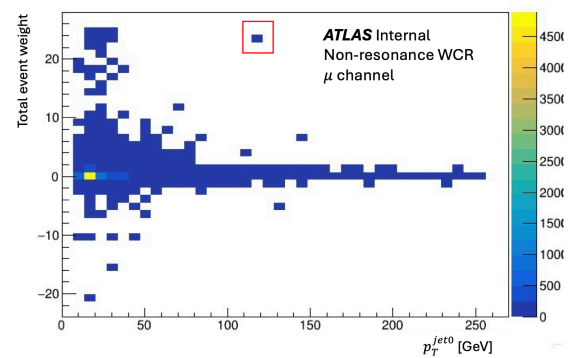
The electron and muon channels give pre-fit data/MC ratios differing by 1.4σ in CRW-Res and 1.7σ in CRW-NonRes. The non-resonant difference is investigated through the corrections applied to simulation, the selection cutflow and the leading-jet p_T distribution. The lepton-dependent efficiency corrections, including the trigger scale factors, are checked separately in the two channels, with no incorrect application identified. The difference becomes visible after imposing $50 < p_T^{j_1} < 250$ GeV.

Figure 8.3 compares the electron and muon distributions separately in data and simulation. The electron data have an upward fluctuation around $p_T^{j_1} = 80\text{--}100$ GeV. The simulated event-weight distributions reveal a second effect: a small number of events carry weights of order 30. Most occur below 50 GeV, but one such event enters the selected muon sample above this threshold, with no corresponding event in the electron sample. A single simulated event can therefore increase the predicted muon yield by several tens of weighted events and lower its data/MC ratio.

These observations are consistent with statistical fluctuations in the electron data and the weighted muon simulation. They do not establish a lepton-dependent correction to the $W + \text{jets}$ rate. The two channels are therefore combined within each CR, and one $W + \text{jets}$ normalisation factor is used for each topology. The extrapolation of this combined constraint is checked in the VRs described in Section 8.5.

(a) Leading-jet p_T in data(b) Leading-jet p_T in simulation

(c) Event weights in the electron simulation



(d) Event weights in the muon simulation

Fig. 8.3 Electron–muon consistency checks in CRW-NonRes. In (a) and (b), red and blue denote the electron and muon channels, and the lower panels show their ratio. Panels (c) and (d) show the simulated event weights against leading-jet p_T . The marked high-weight muon event illustrates the statistical effect discussed in the text.

8.3 Top-quark estimation

The $t\bar{t}$ and single-top backgrounds are constrained in the CRTop regions, which require one τ and an additional electron or muon. The relevant decays include $t\bar{t} \rightarrow b\tau\nu\bar{b}\ell\nu$ and $Wt \rightarrow b\tau\nu\ell\nu$. The additional lepton selects the second leptonic W decay and separates these regions from the lepton-vetoed SRs. Requiring at least one tagged b jet further suppresses $W + \text{jets}$ production. Inclusive $t\bar{t}$ and single-top samples are used. Within the $t\bar{t}$ and Wt samples, the large- E_T^{miss} selection favours decays in which both W bosons decay leptonically. The top-quark purity is approximately 90% after the full CR selections.

The relaxed thresholds in Table 8.4 increase the CR yields. Single-lepton triggers permit the lower $E_T^{\text{miss}} > 100$ GeV threshold. If more than one b jet is selected, the highest- p_T tagged jet defines the topology and the jet- τ observables. The resonant $m_{b\tau}$ requirement is removed.

Table 8.4 Selections for CRTop and VRTop. All regions require $N_\tau = 1$. Single-lepton triggers are used in CRTop and E_T^{miss} triggers in VRTop. The highest- p_T tagged jet is used when $N_b > 1$. Common event-quality and multijet requirements are retained.

Requirement	CRTop		VRTop	
	Res	NonRes	Res	NonRes
N_ℓ		1		0
N_b		≥ 1		2
p_T^ℓ [GeV]		> 28		—
p_T^τ [GeV]		> 100		
p_T^b [GeV]	> 250	(50, 250)	> 250	(50, 250)
E_T^{miss} [GeV]	> 100	> 100	> 200	> 200
m_T [GeV]	> 100	> 100	> 100	> 200
$m_{b\tau}$ [GeV]	—	—	> 400	—
$\Delta\phi(\tau, E_T^{\text{miss}})$	—	> 1.2	—	> 1.2
N_j	≤ 4	≤ 2	≤ 4	≤ 2

The pre-fit distributions in Figures 8.4 and 8.5 show comparable behaviour in the electron and muon channels. The data/MC ratios in Table 8.3 are near 0.8, with no clear dependence on m_T or E_T^{miss} resolved within the statistical precision. The combined top-quark yield is constrained by the fit, while the relative $t\bar{t}$ and single-top contributions are predicted separately in each region. Their ratio is consistent between the SRs and CRs within MC statistical uncertainties. The process-dependent modelling uncertainties are retained, as described in Chapter 9.

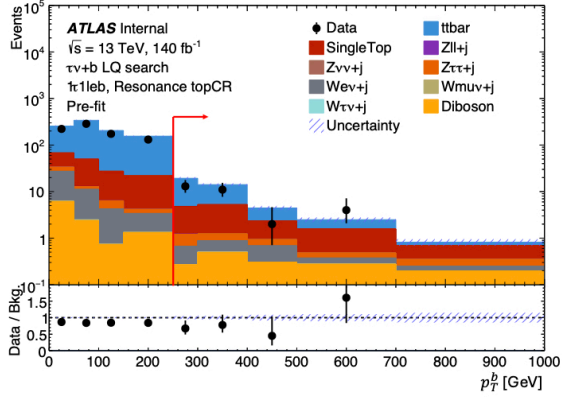
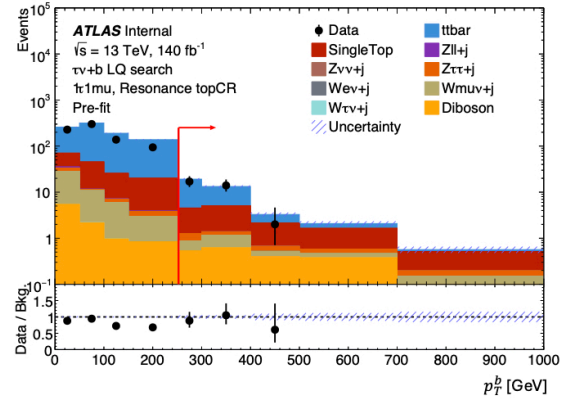
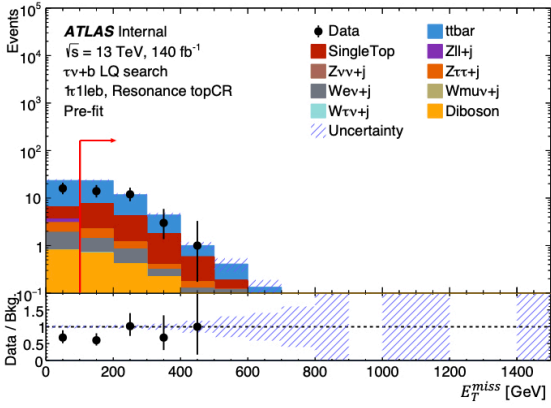
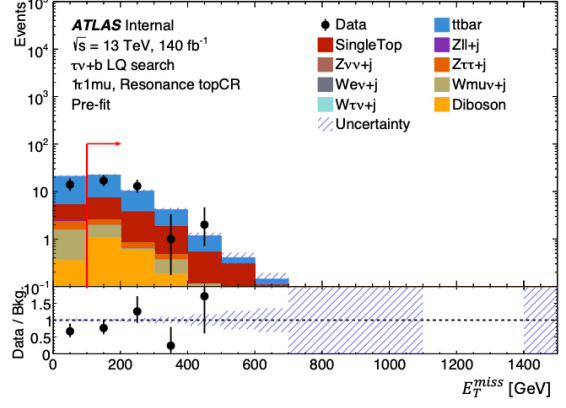
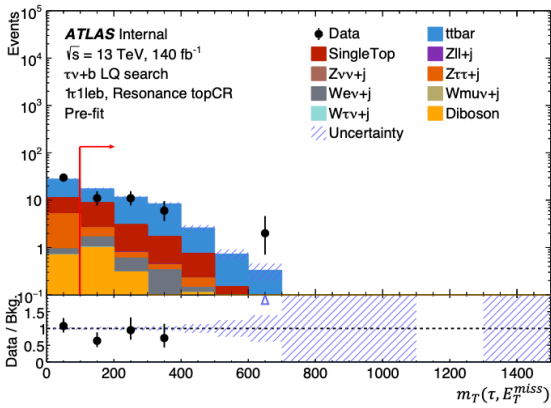
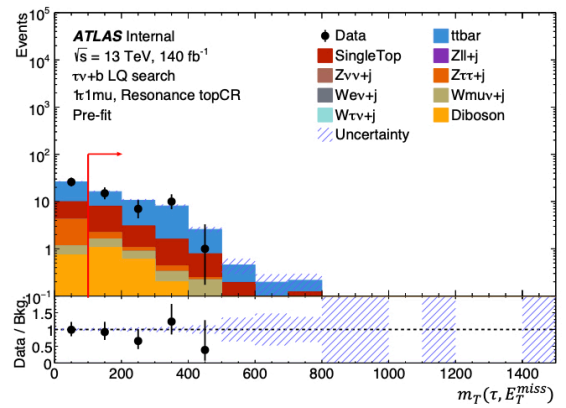
(a) p_T^b , e channel(b) p_T^b , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T , e channel(f) m_T , μ channel

Fig. 8.4 Pre-fit $N - 1$ distributions in CRTop-Res, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

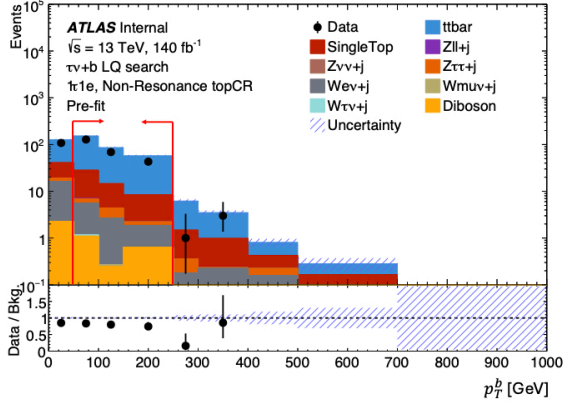
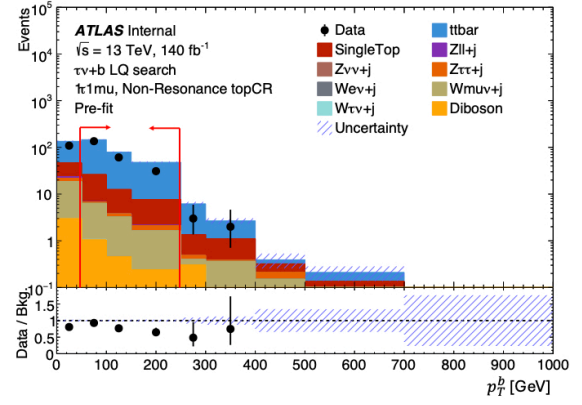
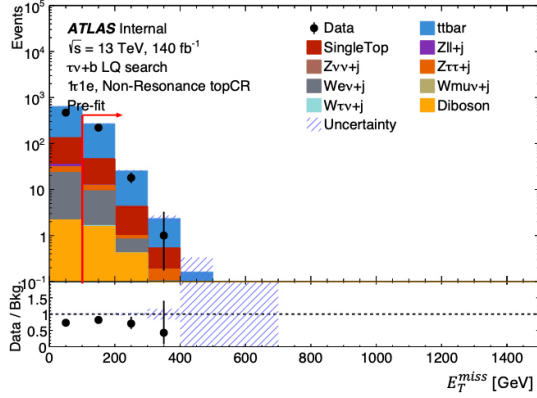
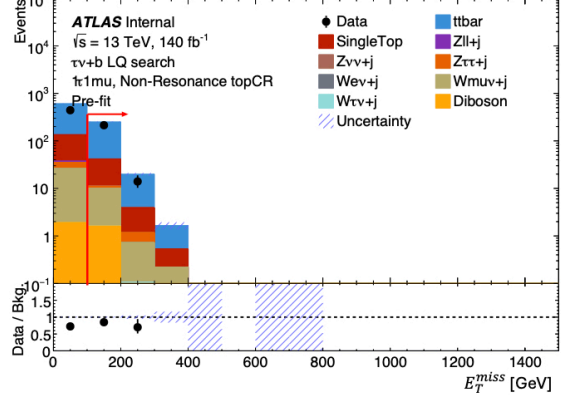
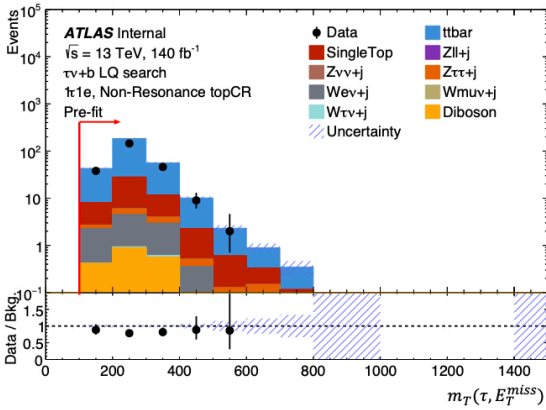
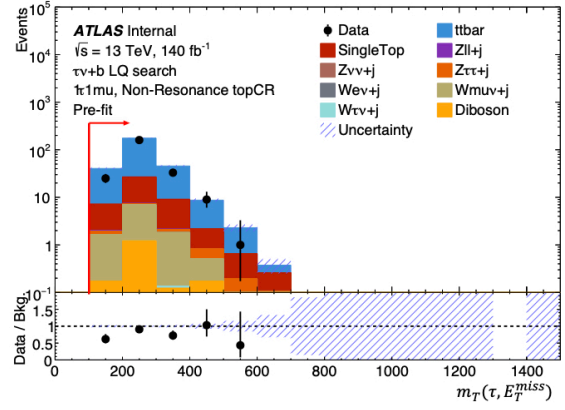
(a) p_T^b , e channel(b) p_T^b , μ channel(c) E_T^{miss} , e channel(d) E_T^{miss} , μ channel(e) m_T , e channel(f) m_T , μ channel

Fig. 8.5 Pre-fit $N - 1$ distributions in CRTop-NonRes, with the electron channel in the left column and the muon channel in the right column. The stacked prediction contains all simulated backgrounds. The lower panels show data divided by the total prediction, and the hatched bands show the statistical uncertainty in that prediction. Arrows indicate the selection boundaries. The last bin includes the overflow.

8.4 Fake- τ estimation

The fake- τ contribution is small because the selected SM background is dominated by events containing a genuine hadronically decaying τ . As discussed in Section 7.2, its principal source at large E_T^{miss} is $Z \rightarrow \nu\nu + \text{jets}$, where a jet passes the τ identification. This contribution is at most about 10% of the background in the SRs considered in the fake-rate study and is much smaller in the one- b -jet SRs. The nominal yield is taken from simulation, and a dedicated $Z \rightarrow \ell\ell$ study determines an uncertainty for the modelling of jets misidentified as τ candidates.

8.4.1 Validation with $Z \rightarrow \ell\ell$ events

The Z validation region (ZVR) selects a visible Z decay together with a τ candidate to test the simulated fake rate in a sample enriched in $Z + \text{jets}$. Two signal electrons or two signal muons are required, with $75 < m_{\ell\ell} < 105$ GeV, together with exactly one τ candidate and a b -jet veto. Events are required to satisfy $1 \leq N_j \leq 3$ and $p_T^\tau > 100$ GeV. Both lepton channels are used to increase the statistical precision. In the muon channel, events are selected using the E_T^{miss} trigger, whereas a single-electron trigger is used in the electron channel.

The quantity $|\vec{p}_T^{\ell\ell} + \vec{p}_T^{\text{miss}}|$ is required to exceed 200 GeV. Adding the reconstructed dilepton momentum to $\vec{p}_T^{\text{miss}}$ treats the visible leptons as unobserved and provides a recoil quantity analogous to the missing momentum in $Z \rightarrow \nu\nu$ events. An additional $p_T^{\ell\ell} > 130$ GeV requirement suppresses top-quark contamination. A transverse-mass threshold of 100 GeV is also applied in this study. The common multijet requirement is retained.

A total of 116 data events pass the ZVR selection, and the simulated $Z + \text{jets}$ purity is approximately 90%. Figure 8.6 shows the p_T^τ distribution with its threshold removed. The simulation overestimates the yield at high p_T^τ . Differences of up to 60% are observed in the overall yields in the validation comparisons. This comparison alone is sensitive to both the fake rate and the production kinematics of $Z + \text{jets}$. The RNN-score distributions in Figure 8.7 provide a further check using looser τ identification and a relaxed transverse-mass requirement. They compare samples with $p_T^\tau > 20$ GeV and $p_T^\tau > 100$ GeV. The better agreement at low RNN scores indicates that the discrepancy is mainly connected to the probability for a jet to pass the τ identification.

The small SR contribution and the precision of this check do not motivate a differential correction of the simulated fake rate. The nominal MC prediction for $Z \rightarrow \nu\nu + \text{jets}$ is therefore retained with an additional 100% normalisation uncertainty. This uncertainty covers the discrepancy observed in the ZVR. Changing the assigned uncertainty from 50% to 100% gives no appreciable change in the tested final results because this background is small. This additional uncertainty describes the fake- τ modelling in this component and is separate from

the uncertainties in genuine- τ reconstruction and identification. No independent fake-factor
prediction is used for the SR yield.

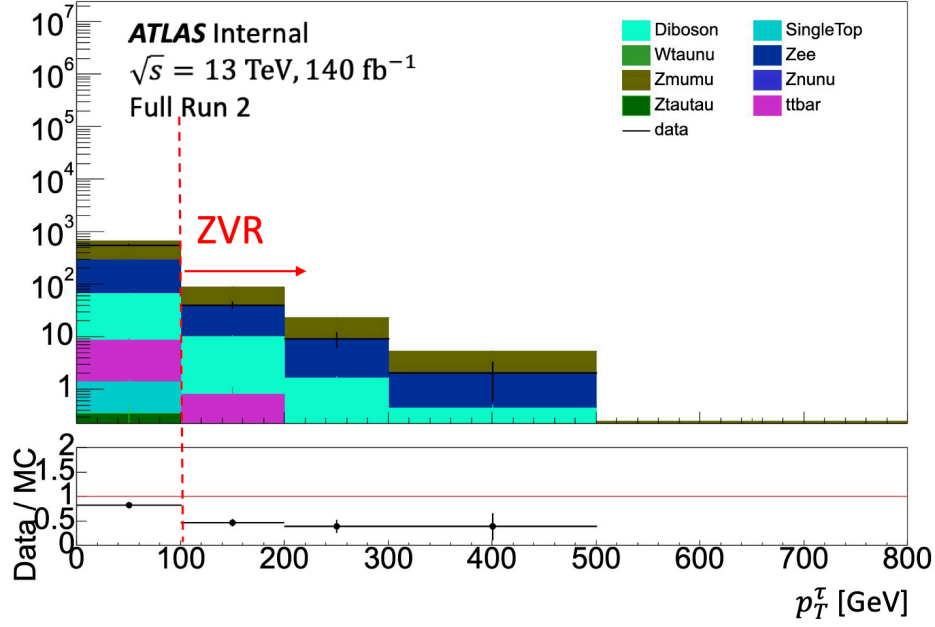


Fig. 8.6 Pre-fit $N - 1$ distribution of p_T^τ in the ZVR. The red arrow indicates the 100 GeV threshold. The lower panel compares data with the total simulation.

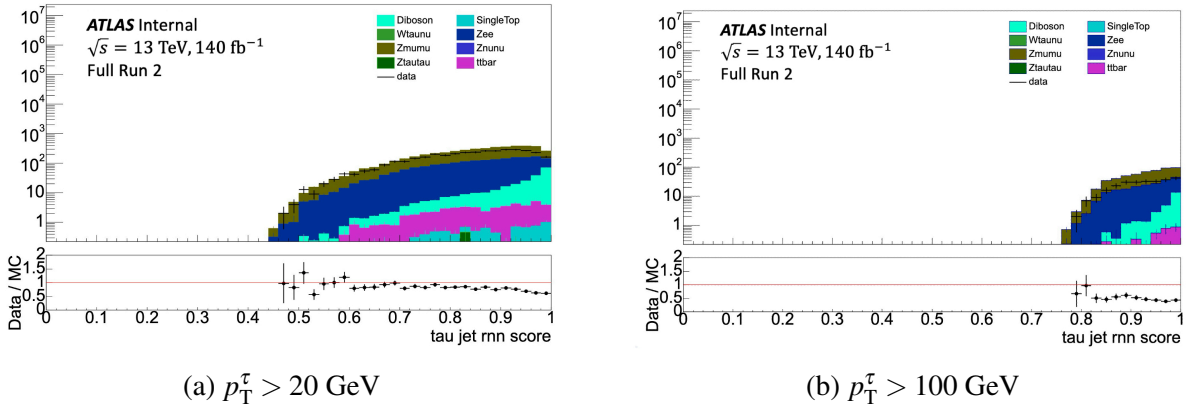


Fig. 8.7 The τ -identification RNN score in $Z + \text{jets}$ -enriched samples with (a) $p_T^\tau > 20$ GeV and (b) $p_T^\tau > 100$ GeV. Looser τ identification and a relaxed transverse-mass requirement are used to compare the low- and high-score regions. These selections differ from the nominal ZVR.

8.5 Validation regions

The VRs test the prediction in selections that isolate the main changes between the CRs and SRs. The nominal VRW and VRTop definitions are given in Tables 8.2 and 8.4. Their pre-fit distributions are used to inspect the simulation before fitting, while their post-fit yields test the extrapolation of the CR constraints.

8.5.1 One- b -jet validation with a light lepton

The VRW regions retain the single-electron or single-muon final state of CRW and require exactly one tagged b jet. The tagged jet replaces the leading untagged jet in the topology and invariant-mass selections. This change tests the prediction for the tagged sample, including the simulated heavy-flavour fraction of $W + \text{jets}$, before replacing the light lepton with a τ in the SR. The top-quark contribution increases under this selection and is predicted using the constraint from CRTop.

The flavour composition of vector-boson-plus-jets production depends on the kinematics and the simulation, as investigated in ATLAS modelling studies and measurements [1–3]. A 30% uncertainty is assigned to the ratio of W production with heavy-flavour hadrons, containing b or c quarks, to production with light-flavour hadrons in the tagged regions. This uncertainty covers possible differences in the flavour composition when transferring the b -vetoed CR constraint to a tagged sample. It is treated as uncorrelated between the resonant and non-resonant categories because they select different jet- p_T ranges. Its implementation and the remaining modelling uncertainties are described in Chapter 9.

The pre-fit $N - 1$ comparisons are shown in Figures 8.8 and 8.9. The electron and muon channels are combined in the plots, and their inclusive yields are listed separately in Table 8.5. The resonant region shows agreement within the statistical precision. The non-resonant region contains more data events than predicted in both lepton channels. The investigation of this excess and the additional check with a τ and one b jet are described below.

Table 8.5 Pre-fit background predictions and observed yields after the complete VRW selections. The prediction uncertainties are statistical only.

Region	Channel	Predicted background	Data
VRW-Res	e	17.0 ± 0.9	20
	μ	16.1 ± 1.0	17
VRW-NonRes	e	22.8 ± 4.4	33
	μ	23.8 ± 3.6	39

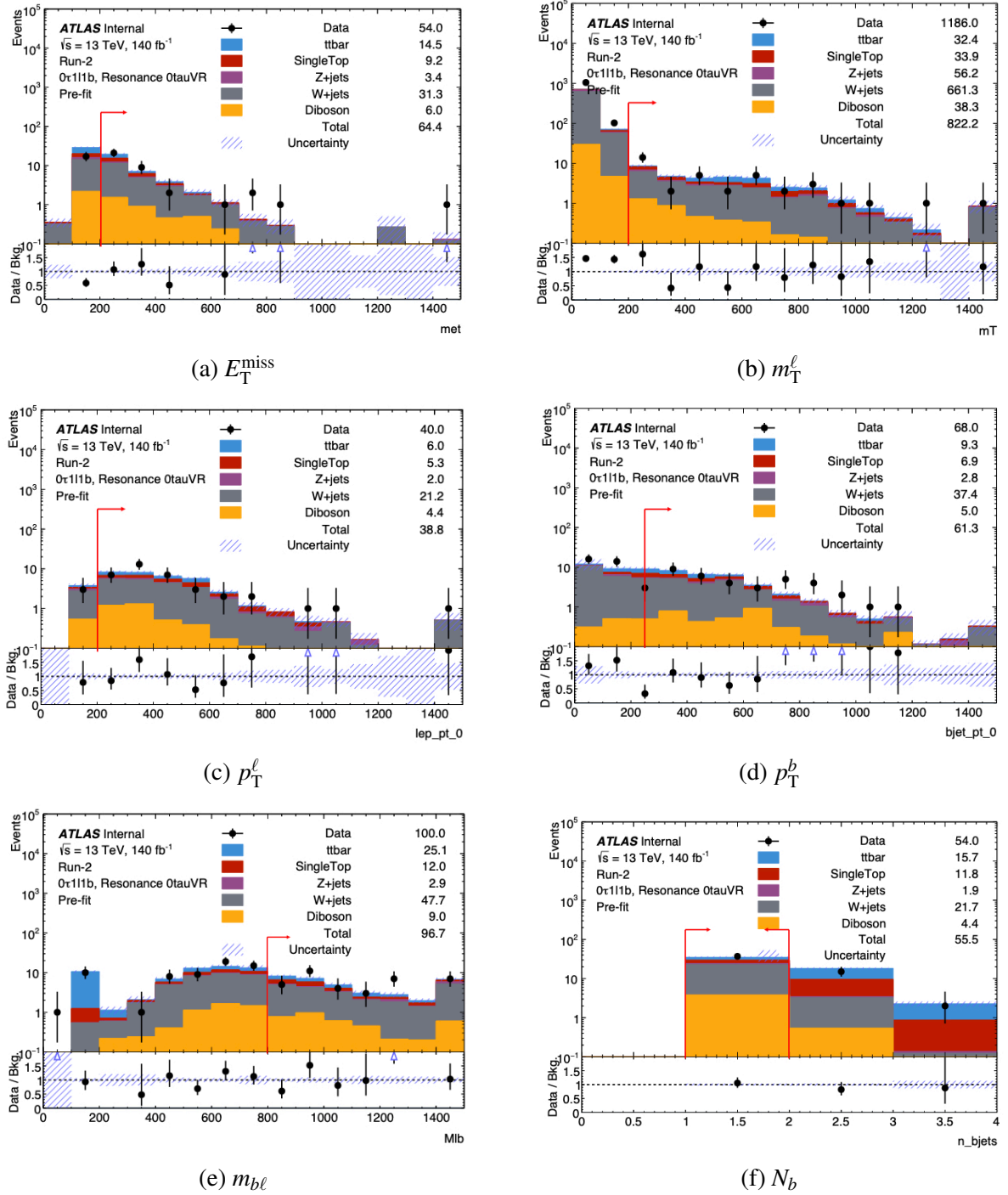


Fig. 8.8 Pre-fit $N - 1$ distributions in VRW-Res. The electron and muon channels are combined. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The original plot labels $0\tau\text{VR}$ denote VRW.

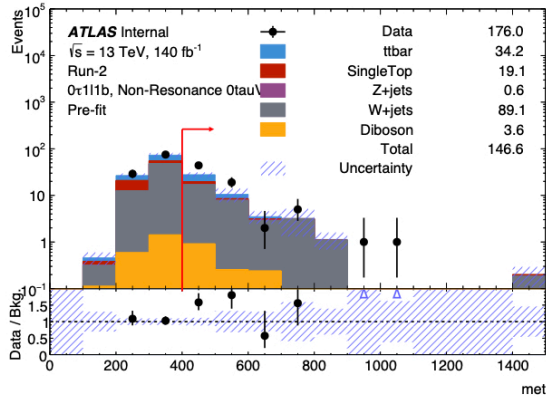
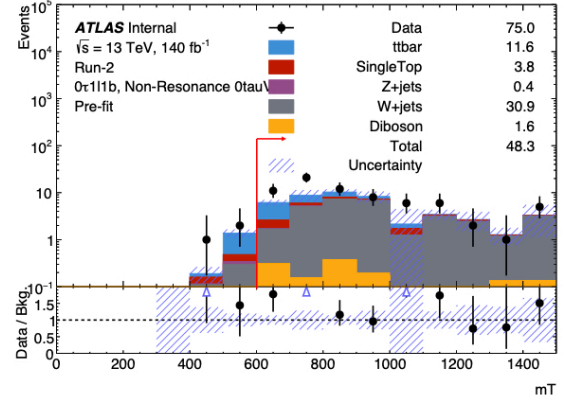
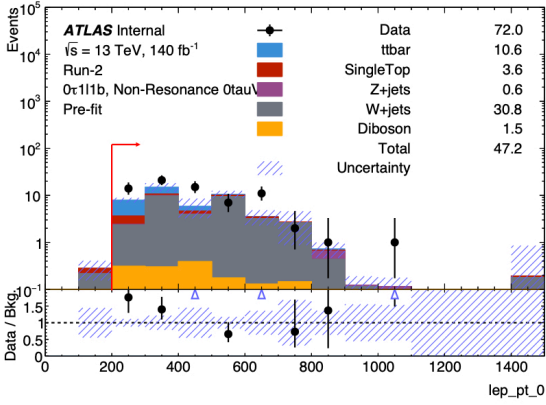
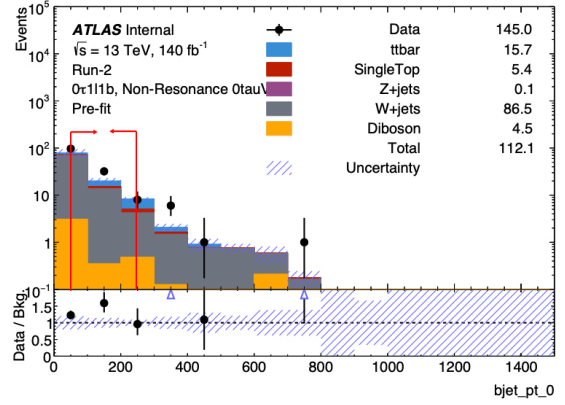
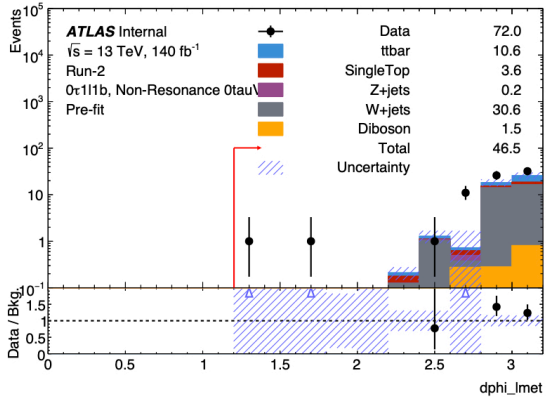
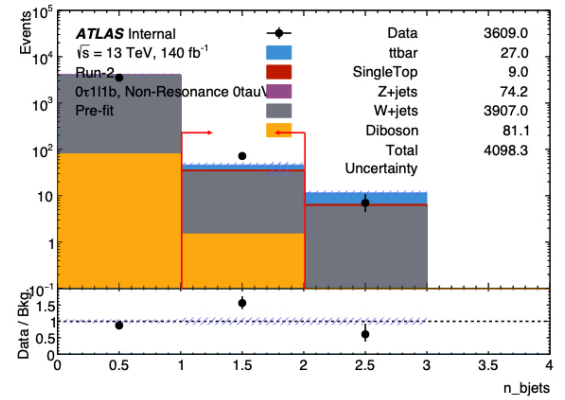
(a) E_T^{miss} (b) m_T^l (c) p_T^l (d) p_T^b (e) $\Delta\phi(\ell, E_T^{\text{miss}})$ (f) N_b

Fig. 8.9 Pre-fit $N - 1$ distributions in VRW-NonRes. The electron and muon channels are combined. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The original plot labels $0\tau\text{VR}$ denote VRW.

8.5.2 Top-quark validation without a light lepton

The VRTop regions require one τ , no electron or muon, and exactly two tagged b jets. The lepton veto reproduces the SR requirement, while the second tag increases the top-quark fraction. The lower p_T^τ and mass thresholds in Table 8.4 provide sufficient events for the check, and $E_T^{\text{miss}} > 200$ GeV keeps the selection on the E_T^{miss} -trigger plateau. The top-quark fractions are approximately 75% and 87% in the resonant and non-resonant regions, respectively.

Figures 8.10 and 8.11 show the pre-fit $N - 1$ distributions. The data are compatible with the simulated distributions within their statistical precision. Single-top production is more prominent in the resonant category, while $t\bar{t}$ dominates the non-resonant category. Together with VRW, these regions test the changes in tagged-jet multiplicity and selected-lepton multiplicity on which the SR prediction depends.

Signal contamination is checked using samples that include both hadronic and leptonic τ decays. The benchmarks are $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.2)$ for the resonant regions and $(2.5 \text{ TeV}, 1.5, 1.0)$ for the non-resonant regions. The contamination is about 1.3% and 0.7% in VRW, and 1.2% and 0.6% in VRTop, respectively.

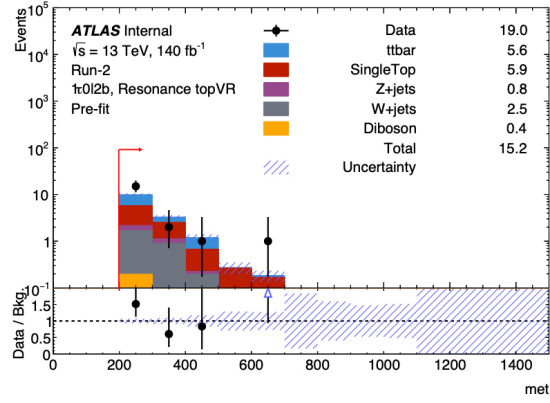
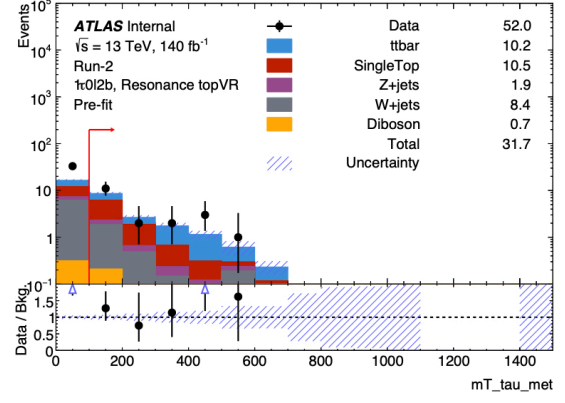
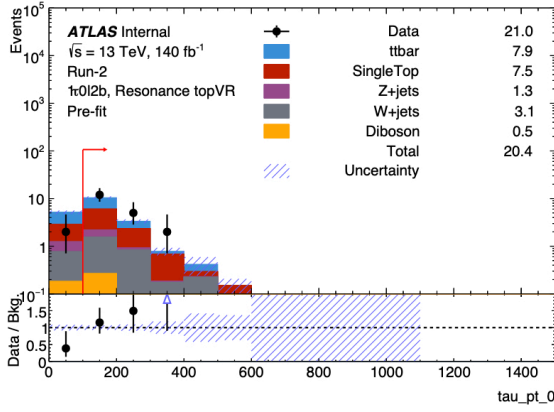
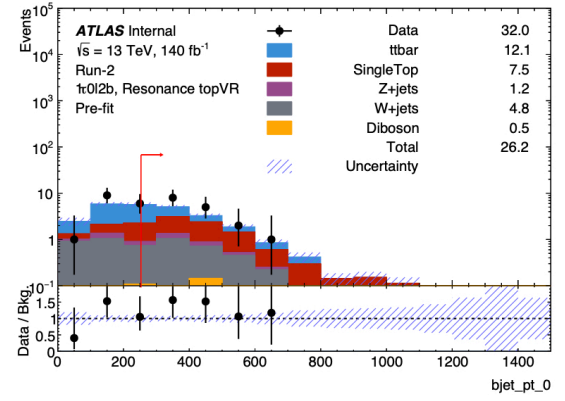
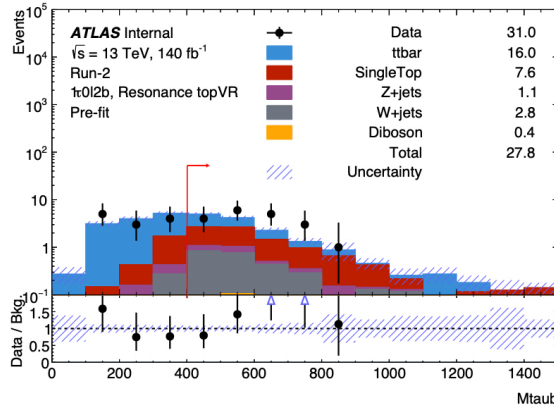
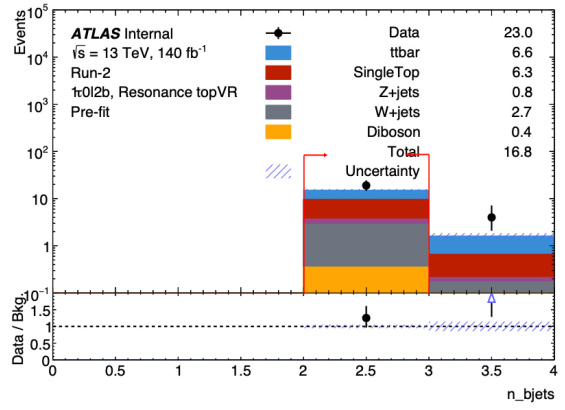
(a) E_T^{miss} (b) m_T (c) p_T^τ (d) p_T^b (e) $m_{b\tau}$ (f) N_b

Fig. 8.10 Pre-fit $N - 1$ distributions in VRTop-Res. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The leading tagged jet is used for the jet-dependent variables.

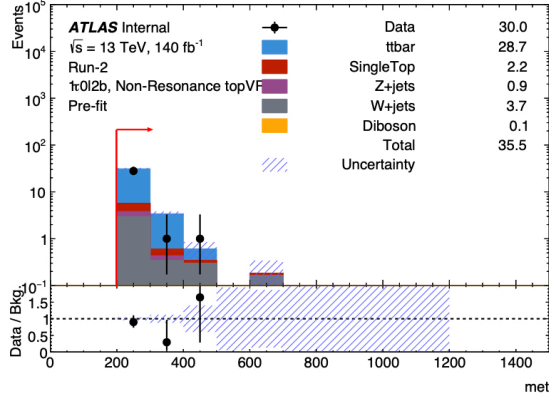
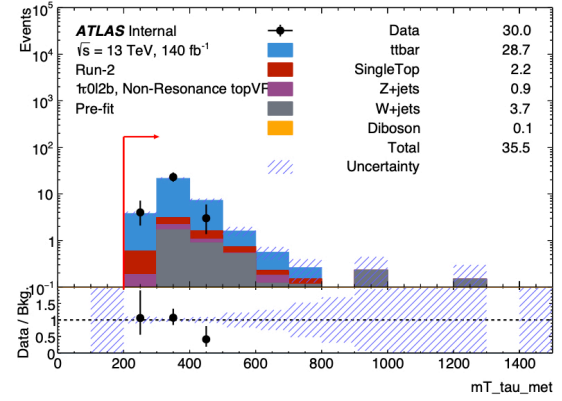
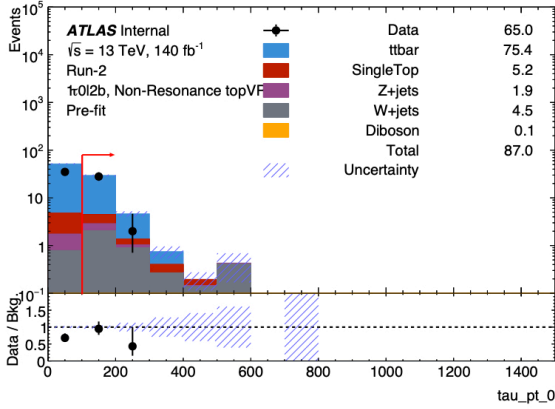
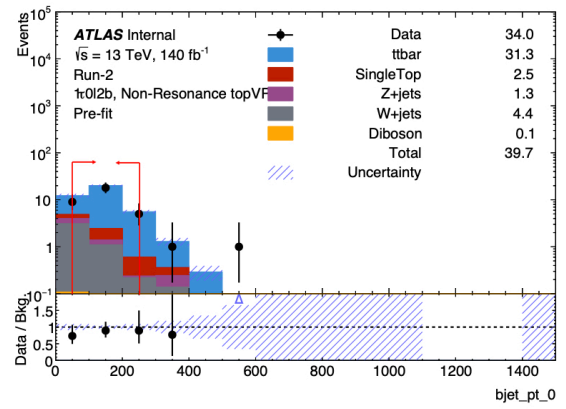
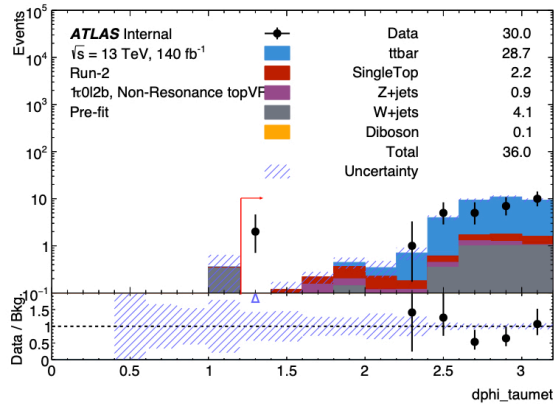
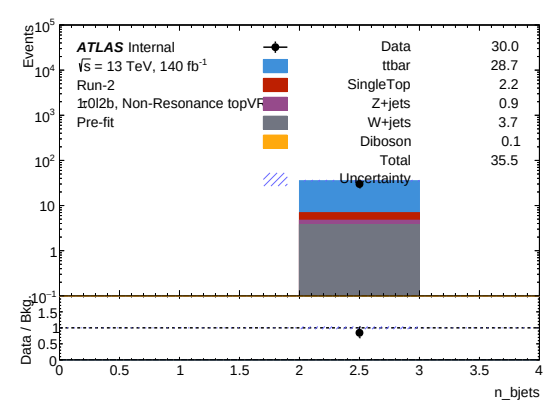
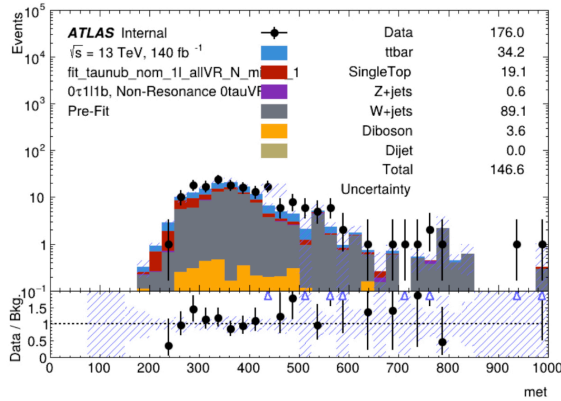
(a) E_T^{miss} (b) m_T (c) p_T^τ (d) p_T^b (e) $\Delta\phi(\tau, E_T^{\text{miss}})$ (f) N_b

Fig. 8.11 Pre-fit $N - 1$ distributions in VRTop-NonRes. The lower panels show data divided by the total simulated background. The hatched bands represent the statistical uncertainty in the prediction. The last bin includes the overflow. The leading tagged jet is used for the jet-dependent variables.

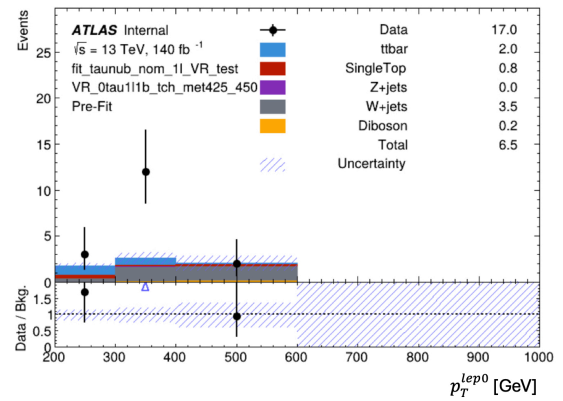
8.5.3 Investigation of VRW-NonRes

The excess in VRW-NonRes remains a deviation of about 2σ after the background-only fit and the assigned uncertainties are included. Its origin is examined through the cutflow, nearby E_T^{miss} selections and the properties of the selected data events. The pre-fit ratio stays close to unity until the $E_T^{\text{miss}} > 400$ GeV requirement is imposed. A finer-binned comparison locates much of the excess in $425 < E_T^{\text{miss}} < 450$ GeV, as shown in Figure 8.12. Changing the E_T^{miss} threshold to 300 GeV or 500 GeV reduces the discrepancy in the statistical-only checks to 1.8σ and 1.5σ , respectively. This behaviour indicates a local excess rather than a persistent offset throughout the high- E_T^{miss} tail.

Within $425 < E_T^{\text{miss}} < 450$ GeV, the excess is concentrated in the $300 < p_T^\ell < 400$ GeV bin, above the 200 GeV lepton threshold. The 12 data events in this interval comprise seven electron and five muon events. Their lepton and b -jet positions show no concentration in a particular detector region. No specific detector problem is identified, and the neighbouring selections give no evidence for a broad failure of the background prediction. The observations are consistent with a statistical fluctuation, although these checks alone do not determine the origin of each excess event.



(a) E_T^{miss} with its threshold removed



(b) p_T^ℓ for $425 < E_T^{\text{miss}} < 450$ GeV

Fig. 8.12 Diagnostic pre-fit distributions for the excess in VRW-NonRes. The finer E_T^{miss} binning in (a) resolves the interval $425 < E_T^{\text{miss}} < 450$ GeV. Panel (b) shows the lepton p_T within that interval. The lower panels show data divided by the simulated prediction.

8.5.4 Additional $1\tau 0\ell 1b$ validation

The additional $1\tau 0\ell 1b$ regions test the background with the same object multiplicities as SR1b in adjacent mass intervals. They provide a direct check of the one- b -jet, hadronic- τ prediction following the excess in the light-lepton validation sample. The resonant region requires $400 < m_{b\tau} < 800$ GeV, and the non-resonant region requires $500 < m_T < 600$ GeV. These upper bounds keep the regions separate from the corresponding SRs. Further reductions in p_T^τ , E_T^{miss} and m_T increase the event yields, as summarised in Table 8.6. All other requirements follow the corresponding SR1b selections.

Table 8.6 Selections and yields in the supplementary $1\tau 0\ell 1b$ validation regions. Both use E_T^{miss} triggers and the remaining SR1b requirements. The background-only fit to the CRs is extrapolated to these regions. The quoted prediction uncertainties are those of this validation study, without an additional uncertainty for the changed mass intervals.

	Resonant	Non-resonant
p_T^τ [GeV]	> 100	> 100
E_T^{miss} [GeV]	> 200	> 200
m_T [GeV]	> 100	$500 < m_T < 600$
$m_{b\tau}$ [GeV]	$400 < m_{b\tau} < 800$	No requirement
Signal contamination	$\sim 2\%$	$\sim 7\%$
Post-fit background	25.4 ± 1.5	8.0 ± 1.7
Observed data	24	9

Signal contamination is evaluated before examining the data, using the same benchmarks as in Section 8.5.2. The estimated contributions are about 2% and 7% in the resonant and non-resonant regions, respectively. Background yields of 25.4 ± 1.5 and 8.0 ± 1.7 events are predicted after the CR fit, compared with 24 and 9 observed events. The agreement supports the transfer of the background normalisation to the $1\tau 0\ell 1b$ final state in these lower-mass intervals. No additional uncertainty for the changed $m_{b\tau}$ or m_T selections is included in this supplementary study, so its agreement is a cross-check rather than a separate determination of the uncertainty in the high-mass SR prediction.

References

- [1] ATLAS Collaboration. Modelling and computational improvements to the simulation of single vector-boson plus jet processes for the ATLAS experiment. *JHEP*, 08:089, 2022. doi: 10.1007/JHEP08(2022)089.
- [2] ATLAS Collaboration. Measurements of the production cross-section for a Z boson in association with b - or c -jets in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Eur. Phys. J. C*, 84:984, 2024. doi: 10.1140/epjc/s10052-024-13159-w.
- [3] ATLAS Collaboration. Measurements of WH and ZH production with Higgs boson decays into bottom quarks and direct constraints on the charm Yukawa coupling in 13 TeV pp collisions with the ATLAS detector. *JHEP*, 04:075, 2025. doi: 10.1007/JHEP04(2025)075.